

User Guide

Prior Authorization Multi-Agent Platform

**DEMONSTRATION SYSTEM — SYNTHETIC DATA — NOT FOR
CLINICAL
OR COVERAGE USE**

Document type: User Guide — features, workflow, and benefits

Version 1.1

Audience: clinical reviewers, providers/administrators, and evaluators of the platform

1. Welcome

This platform is a demonstration of how a prior authorization (PA) request can move from submission to a fully explained, evidence-backed determination — with a real human always making the final call. This guide walks through every screen, explains what each feature does, and why it matters.

Every patient and every payer policy in this system is synthetic. Nothing here is a real coverage decision, and the platform says so on every screen. It exists to demonstrate the workflow, not to make real determinations.

2. What This Platform Gives You

Feature	Benefit
Structured request intake	No more free-text guesswork — requested service, diagnosis codes, urgency, and prior treatments are captured explicitly, up front.
Live multi-agent pipeline	Watch each step of the evaluation happen in real time instead of waiting on a single opaque spinner — you see exactly what the system is doing and when.
Criterion-by-criterion evidence	Every individual policy rule is shown as MET, NOT MET, or UNKNOWN with the specific patient fact that justified it — never a black-box "denied".
Deterministic, reproducible decisions	The same request always produces the same determination — the outcome is a rules engine, not a guess, so it can be explained and defended.
AI-assisted plain-language rationale	A clear paragraph explaining the decision, optionally drafted by Anthropic Claude or OpenAI (whichever provider is configured) — but the AI never makes the decision itself, only explains it.
Mandatory human review	Nothing is final until a clinical reviewer signs off — every determination, whether approved, denied, or pending, goes through this gate.
Full audit trail	Every step, every criterion, every reviewer decision is recorded and can be downloaded as a PDF report at any time.
Operational visibility	A dashboard shows volume, outcomes, how often reviewers override the AI, per-agent performance, and (if enabled) real AI cost — filterable to the last 7 days, 30 days, 1 month, or 1 year, all in one place.

3. Getting Started

Once the platform is running (see the project README for setup), open the frontend in your browser. You will see four sections in the top navigation:

- **Patients** — browse synthetic patients and submit new requests.
- **Review Queue** — determinations waiting for a clinical reviewer's sign-off.
- **Policy Search** — search the payer policy library directly.
- **Observability** — system health and operational metrics.

4. Submitting a Prior Authorization Request

1. From the **Patients** page, click a patient card to open their record.
2. Review the patient's demographics, vitals/labs, history, medications, and allergies at the top of the page.
3. Scroll to **New prior authorization request** and fill in:

Field	What to enter
Requested service *	The procedure, medication, or equipment being requested (e.g. "Bariatric surgery", "Lumbar spine MRI"). Required.
Service category	imaging / surgery / medication / therapy / dme / other.
Urgency	routine / urgent / emergent.
Diagnosis codes	Comma-separated ICD-10 codes or plain descriptions. Auto-filled from the patient's documented history when you open their record (e.g. a patient with "type_2_diabetes, hypertension" pre-fills "E11.9, I10") — add or remove codes to match this specific request.
Prior treatments tried	Comma-separated list of what's already been attempted (e.g. "physical therapy for 8 weeks").
Additional clinical justification	Optional free-text context for the reviewer.

1. Click **Submit for determination**.

The Requested service field is required. If you submit without it, the field is outlined in red and focused automatically so you can fill it in immediately — nothing is lost.

5. Watching the Multi-Agent Pipeline

After you submit, a **Multi-agent orchestration** panel appears and lights up step by step, live, as each of the six agents actually finishes its work on the server — this is not a simulated animation. Below it, a growing **Live agent details** log shows the exact input and output each agent produced, the moment it produced it.

Step	In plain terms
Clinical Intake	Checks the patient record and the request are complete enough to proceed.
Guideline Retrieval	Finds the one payer policy document that actually applies to this request.
Risk Stratification	Summarizes the patient's overall clinical risk factors for context.
Criteria Matching	Checks the request against every individual rule in that policy, one at a time.
Determination	Decides approve / deny / pend from the criteria results, and drafts a plain-language explanation.
Safety Validation	Double-checks the decision and evidence before handing it off for human review.

If information is critically incomplete, the pipeline stops right after Clinical Intake and returns a "pending" result asking for more information, rather than guessing.

6. Understanding Your Determination

Once the pipeline finishes, a result card appears with:

- **Determination badge** — Approved (green), Denied (red), or Pended (amber, needs more information or human judgment).
- **Rationale** — a plain-language explanation, labeled as either AI-drafted (with the provider, model name, and token count) or from the deterministic template (when no AI provider is configured).
- **Risk factors & identified conditions** — clinical context pulled from the patient's record.
- **Criteria evaluated** — every policy rule that was checked, each tagged MET, NOT MET, or UNKNOWN, with the specific patient evidence behind that tag.
- **Retrieved policy evidence** — the actual policy text the determination is grounded in, with a relevance score.
- **Safety flags** — anything the system wants a human to pay particular attention to (e.g. an allergy/medication overlap, high risk, or evidence it couldn't fully verify).

Every determination — even an Approved one — is explicitly marked as requiring clinical reviewer sign-off. A banner links straight to that request's entry in the Review Queue.

7. Reviewing a Determination (Human-in-the-Loop)

This is the safety net at the center of the platform: no determination is ever presented as final without a person confirming it.

1. Open the **Review Queue** from the top navigation. Every completed determination awaiting sign-off is listed, oldest first, with the patient, requested service, and the AI's determination.
2. Click a card to open its full detail — the complete determination (exactly as shown in step 6) alongside a review form.
3. Enter your name.
4. Choose **Uphold** to accept the AI's determination as-is, or **Override** to replace it.
5. If overriding, you must select the correct final determination and write a note explaining why — this is enforced, not optional, so every override is accountable.
6. Click submit. The request leaves the queue and is now final.

Once reviewed, the reviewer's name, decision, final determination, and notes are permanently attached to that request and visible on its detail page and in its PDF report.

8. Downloading a PDF Report

From either the request's result screen or its review page, click **Download PDF report**. The PDF includes the full determination summary, rationale, every criterion evaluated with its evidence, the retrieved policy text it's grounded in, any safety flags, and — once reviewed — the reviewer's name, decision, and notes. It is a rendering of exactly what was decided and recorded, suitable for sharing or filing.

9. The Observability Dashboard

The Observability tab answers "is the system healthy, and is it working well?" at a glance.

A period selector next to the page title — **Last 7 days**, **Last 30 days**, **Last 1 month**, **Last 1 year**, or **All time** — scopes system health, volume & value, review stats, per-agent stats, and recent requests to that trailing window.

Section	What it tells you
System health	Whether the database, the policy search index, and the AI integration (and which provider — Anthropic or OpenAI — it's using) are up.
Volume & value	How many requests have come in, how many completed, average turnaround time, and the approve/deny/pend split as a chart.
Human-in-the-loop review	How many requests are waiting for review, how many have been reviewed, and the override rate — how often a human actually disagreed with the AI. This is a key trust indicator.
Per-agent latency	How long each of the six agents takes on average and at the slow end (p95), on a chart, so a slow step is easy to spot.
Per-agent reliability	How often each agent succeeds versus errors, per agent.
LLM usage & cost	If a provider is configured: how many times it was called, how many tokens it used, and an estimated cost. If not configured, it clearly states that a deterministic template is being used instead, and names which environment variable (ANTHROPIC_API_KEY or OPENAI_API_KEY) to set.
Cost per execution	A separate filter (Last 7 days / Last 30 days / This month / This year / Custom range) drives a spend trend chart and a per-run cost breakdown, independent of the page-wide period selector above.
Recent requests	A quick table of the latest activity within the selected period.

10. Searching the Policy Library

The **Policy Search** tab lets you search the payer policy corpus directly — useful for understanding what criteria exist for a given service before submitting a request. Enter a query (e.g. "bariatric surgery BMI criteria"), choose how many results to return, and the matching policy sections appear with their relevance score and source document.

11. Frequently Asked Questions

Why did my request come back "pending" instead of approved or denied?

Either critical information was missing from the request/patient record, or at least one policy criterion could not be confidently evaluated (UNKNOWN) — both route to a human for judgment rather than a guess.

Can the AI model deny a request on its own?

No. The approve/deny/pend decision is always produced by a deterministic rules engine from the evaluated criteria. The AI model, when configured, only writes the explanation for a decision that has already been made.

Which AI provider does this use?

Either Anthropic Claude or OpenAI — selected by the LLM_PROVIDER setting. Only the selected provider's API key needs to be configured; both are used purely to phrase an already-decided outcome, never to make it.

What happens if I try to review the same request twice?

The system rejects it — a request can only be reviewed once. If a mistake was made, it should be corrected through your organization's normal correction process, not a second review submission.

Why do I have to enter notes when I override?

So every disagreement with the AI is accountable and explained — this is enforced by the system, not left optional.

Is any of this real patient or policy data?

No. Every patient and every policy document in this system is synthetic and clearly labeled as demonstration data throughout the platform and in every generated document.

12. Glossary

Term	Meaning
Determination	The approve / deny / pend outcome for a request.
Criterion	One individual rule from a payer policy (e.g. a BMI threshold).
MET / NOT MET / UNKNOWN	Whether a criterion is satisfied, not satisfied, or cannot be confidently evaluated from the information given.
Uphold	A reviewer accepts the AI's determination as final.
Override	A reviewer replaces the AI's determination with their own, with required notes.
Override rate	How often reviewers override the AI — a trust/quality signal shown on the dashboard.
Rationale	The plain-language explanation of why a determination was reached.
LLM provider	Which AI vendor drafts the rationale/scenario text — Anthropic Claude or OpenAI, selected via the LLM_PROVIDER setting.